Problem-Solution Fit

Date	03-07-2026
Team ID	SWTID-2026-5714
Project Name	OptiCrop: Smart Agriculture Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:
Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and

1. CUSTOMER SEGMENT(S) [CS]	**2. PROBLEMS / PAINS + ITS FREQUENCY [PR]**	**3. TRIGGERS TO ACT [TR]**
Primary: Small & marginal farmers in AP/TS with 1-5 acre land Secondary: FPOs, Agriculture officers, Agri-input dealers Tertiary: Govt policymakers for subsidy planning	P1: Wrong crop selection due to lack of scientific guidance P2: Low agricultural productivity caused by unsuitable crop choices P3:Difficulty selecting crops using soil nutrient values P4: Lack of simple AI-based decision support for farmers	T1: Monsoon onset — Need to decide crop in 7 days T2: Previous season loss — "this time must not repeat the mistake" mindset T3: Neighbor got good price — FOMO T4: Govt subsidy announcement for specific crop

limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

4. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL]	7. BEHAVIOR + ITS INTENSITY [BE]	8. CHANNELS OF BEHAVIOR [CH]
Budget: Can't afford ₹5000+ paid apps Devices: 60% use feature phones, not smartphones Literacy: 40% can't read English, low digital skills Network: Poor internet in villages	High Intensity: Call agriculture officer/AEO for advice - 80% farmers do this weekly Medium: Watch TV programs like Annadata - 60% farmers High: Ask	ONLINE Agriculture websites Mobile internet YouTube farming channels OFFLINE • Agriculture officers • Local farmers • Village agriculture meetings
	neighbor farmers what they planted - 90% farmers Result: Herd behavior, market glut	for introduction

5. AVAILABLE SOLUTIONS PROS & CONS [AS]

Pros: Government agriculture websites provide general farming information.

Existing crop recommendation applications offer basic guidance.

Cons: No personalized AI-based crop prediction.

Limited use of soil nutrient parameters.
Complex interfaces for many farmers.

9. PROBLEM ROOT / CAUSE [RC]

R1: Decisions based on tradition/neighbors, not data

R2: No single platform combining soil + weather + market R3: Existing solutions not vernacular + not SMS friendly

R4: Illiteracy & tech phobia

10. YOUR SOLUTION [SL]

OptiCrop AI System

- Farmers enter soil nutrient values (N, P, K), temperature, humidity, pH, and rainfall through a simple web interface.
- A trained **Random Forest** machine learning model predicts the most suitable crop.
- The Flask-based web application provides Home, About, Predict Crop, and Result pages.
- The solution is lightweight, easy to use, and can be deployed online for public access.
- Future enhancements include weather integration, fertilizer recommendation, and crop disease prediction.

6. EMOTIONS BEFORE / AFTER [EM]

BEFORE — Anxiety: "Will the rains fail and ruin the crop? How will I repay the loan?"

AFTER — Confidence: "The AI model recommended a suitable crop based on my soil and climate conditions."